

Alex Siegel

(818) 439-7884 | alexrigelsiegel@gmail.com | Pasadena, CA 91125
<https://alexrsiegel.github.io> | <https://www.linkedin.com/in/alex-siegel>

SUMMARY

I'm a biophysicist with 15 years of experience studying the mechanisms of proteins. My recent work uncovers a detailed mechanistic understanding of how protein chaperones preserve their misfolding clients. I'm interested in understanding the molecular mechanisms of proteins that underlie diseases and applying this knowledge to develop novel therapeutics.

RESEARCH EXPERIENCE

* *Equal contribution first author publications.*

Postdoctoral Scholar in Chemistry

2017-present

California Institute of Technology, Pasadena, CA | Prof. Shu-ou Shan

- Developed a pipeline to map transient protein-protein interactions between a chaperone and its clients by NMR, EPR, and crosslinking-coupled mass spectrometry ([McAvoy, Siegel, et al, JBC, 2018](#); [Siegel*, McAvoy*, et al, JMB, 2020](#)).
- Expert review of EPR applications to the study of chaperones and aggregation ([Siegel*, Singh*, et al, RSC, 2023](#)).
- Discovered a chaperone critical for photosynthesis also protects chlorophyll-synthesizing enzymes ([Wang et al, PNAS, 2018](#)) and how its heat protection of these clients is regulated ([Ji*, Siegel* et al, Nature Plants, 2021](#)). Provides potential target to bioengineer heat-tolerant plants.
- Uncovered novel thermostat-like chaperone activation mechanism triggered by a disorder transition ([Siegel*, McAvoy*, et al, JMB, 2020](#)) and demonstrated this targets different native clients ([Siegel et al, Science Advances, 2025](#)) or non-native amyloid-forming clients ([Prov. Patent](#)). Enables bioengineering chaperones against toxic amyloids.
- Solved cryo-EM structure of a human mitochondrial chaperone, revealing two assembly states: one to pull clients out of aggregates and another to protect their refolding ([Gupta*, Lentzsch*, Siegel*, et al, Science Advances, 2023](#)). Aids search for targeted therapeutics against several congenital mitochondrial diseases.

Graduate Researcher in Biophysics

2009-2016

University of California, Berkeley | Prof. David Wemmer

- Found mechanistic insights into stress-responsive bacterial transcription factors that apply force to RNA polymerase to open DNA and initiate mRNA transcription ([Siegel et al, JMB, 2016](#)). Provides new potential antibiotic targets.
- Uncovered effects of force propagation on the conformation of the RNAP polymerase subunits during its transcriptional activation by molecular tweezers ([Siegel, Dissertation, 2016](#)).
- Monitored effects of pressure-induced conformational changes to piezophilic transcription factors by high pressure NMR.

SKILLS

- *Independent researcher.* Initiated multiple original projects leading to high impact first author publications.
- *Science communicator.* Presented at bi-monthly seminars and 5+ large conferences; drafted 8 manuscripts.
- *Collaborative worker.* Initiated and managed 5 inter-lab collaborations leading to published interdisciplinary work.
- *Experienced mentor.* Directly trained 20 students in first biochemistry projects; led research teams; taught 6 classes.
- *Quick learner.* Broad experience in biophysics, biochemistry, and molecular biology techniques (below).

Biophysics & Structural Biology. • Protein NMR (15 yrs) • Bulk and single molecule FRET / anisotropy (5 yrs) • Cross-linking coupled mass spectrometry (3 yrs) • Circular dichroism (3 yrs) • Cryo-EM (2yrs) • Mass photometry (2 yrs) • X-ray crystallography (1 yr) • EPR (1 yr) • Molecular dynamics (1 yr) • Molecular tweezers • ITC • MST • SAXS

Molecular Biology & Biochemistry. • Cloning / expression / protein purification (15 yrs) • *in vitro* translation (5 yrs) • Amber suppression (5 yrs) • Immunoassays / western blotting / immunoprecipitation (8 yrs) • Fluorescent labeling (5 yrs) • Chemical crosslinking (3 yrs) • Small molecule purification / HPLC • SEC-MALS • Light scattering (8 yrs) • High-throughput screening by microplate reader

Data analysis & software. • Python • R • Matlab • Graphpad • NMR (*MestrelNova*, *Topspin*, *NMRPipe*, *NMRView*, *CARA*) • Cryo-EM (*CryoSPARC*, *Relion*) • Mass spectrometry (*ProteinProspector*, *MeroX*) • Structure determination (*PHENIX*, *Coot*) • Structure visualization (*ChimeraX*, *Pymol*) • Structure prediction (*Gromacs*, *AlphaFold*)

EDUCATION

University of California, Berkeley – Ph.D., Biophysics (with Chemical Biology specialization)

2016

Thesis: *Mechanisms of σ^{54} bacterial transcription activation* | Advisor: Prof. David E. Wemmer

California Institute of Technology – B.S., Biology & Chemistry (Double major, with Honors)

2008

PUBLICATIONS & PATENTS

*Indicates equal contribution first author publications.

- **Siegel A**, Kroon G, Zhao CQ, Wang P, Wright PE, and Shan SO. *Switchable client specificity in a dual functional chaperone coordinates light harvesting complex biogenesis*. [Science Advances](#) (2025). In Press.
- **Siegel A***, Singh J*, Qin PZ & Shan SO. *EPR Studies of Chaperone Interactions and Dynamics*. Biophysics of Molecular Chaperones (eds. Hiller, S., Liu, M. & He, L.) 242–277. [Royal Society of Chemistry](#) (2023).
- Gupta, A*, Lentzsch AM*, **Siegel A***, Yu Z, Chio US, Cheng Y, & Shan SO. *Dodecamer assembly of a metazoan AAA+ chaperone couples substrate extraction to refolding*. [Science Advances](#). 9(19), eadf5336 (2023).
- Patent ID: CIT-9049-P. Provisional patent application for use of chaperones against neurodegenerative diseases. (2023)
- Ji S*, **Siegel A***, Shan S., Grimm B. & Wang P. *Chloroplast SRP43 autonomously protects chlorophyll biosynthesis proteins against heat shock*. [Nature Plants](#). 7, 1420-1432 (2021).
- **Siegel A***, McAvoy C*, Lam V, Liang FC, Kroon G, Miaou E, Griffin P, Wright P, Shan SO. *A Disorder-to-Order Transition Activates an ATP-Independent Membrane Protein Chaperone*. [Journal of Molecular Biology](#). 432, 166708 (2020).
- McAvoy C, **Siegel A**, Piszkiwicz S, Miaou E, Yu M, Nguyen T, Moradian A, Sweredoski MJ, Hess S, Shan SO. *Two Distinct Sites of client protein interaction with the chaperone cpSRP43*. [Journal of Biological Chemistry](#). RA118, (2018).
- Wang P, Liang FC, Wittmann D, **Siegel A**, Shan SO, Grimm B. *Chloroplast SRP43 acts as a chaperone for glutamyl-tRNA reductase, the rate-limiting enzyme in tetrapyrrole biosynthesis*. [Proceedings of the National Academy of Sciences](#). 115(15), E3588-96 (2018).
- **Siegel AR**, Wemmer DE. *Role of the σ^{54} Activator Interacting Domain in Bacterial Transcription Initiation*. [Journal of Molecular Biology](#). 428(23), 4669-85 (2016).
- **Siegel AR**, *Mechanisms of σ^{54} bacterial transcription activation*. Doctoral Dissertation University of California, Berkeley. Publication No. 10192026. [ProQuest Dissertations Publishing](#) (2016).

SELECTED PRESENTATIONS

- **33rd Western Photosynthesis Conference**. “Regulated capture and delivery of the light harvesting complex proteins by chloroplast SRP”, Biosphere 2, AZ (2024).
- **ASBMB annual meeting in Experimental Biology: Spotlight Session**, “Client Specificity of an ATP-independent Chaperone is Regulated by a Temperature Sensitive Switch”, Philadelphia, PA (2022).
- **ASBMB Science in a Flash: Flash Talk**, “A molecular thermostat that protects plants from heat”, Virtual (2022).
- **Center for Molecular and Cellular Medicine**, “A molecular thermostat switches on alternative chaperone activity during heat stress”, Caltech (2021).
- **EMBO workshop: New challenges in protein translocation across membranes**, “A disorder-to-order transition activates the chloroplast signal recognition particle”, Virtual Meeting (2021).
- **Center for Molecular and Cellular Medicine**, “Regulated substrate capture and release by a membrane protein chaperone”, Caltech (2020).
- **Gordon Research Conference: Computational Aspects of Biomolecular NMR**, “Role of the σ^{54} Activator-Interacting Domain in Bacterial Transcription Initiation”, Lucca, Italy (2015).
- **Structure Supergroup Biophysics Seminar Series**, “Mechanisms of σ^{54} Bacterial Transcription Initiation”, UC Berkeley (2012 & 2014).
- **Biophysics Annual Retreat**, “Mechanisms of σ^{54} Bacterial Transcription Initiation”, UC Berkeley (2010 & 2014).

LEADERSHIP & SERVICE

- **Caltech Postdoctoral Association - Secretary and Vice Chair** 2018-2022
Secured more career development funding, planned dozens of career and social events including LA-wide ExploreCaltech, improved onboarding of new postdocs, coordinated postdocs across diverse academic divisions.
- **ISEF Judge** 2017
Judge for biochemistry and biophysics at the Regeneron International Science and Engineering Fair.
- **UC Berkeley Graduate Assembly - Delegate** 2011-2012
Represented the Biophysics group, advocated department issues on behalf of other graduate students.
- **Caltech Academic Research Council - Secretary** 2005-2007
Gathered course feedback, liaised with faculty to implement curriculum and teaching improvements.